COMS4732 — Computer Vision 2

Professor Aleksander Holynski

HW3: Simple Structure from Motion

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Overview

This assignment implements a two-view calibrated SfM pipeline on the staff-provided image pair: camera intrinsics estimation, SIFT feature detection, NNDR feature matching, RANSAC pose estimation via the Essential matrix (8-point), and sparse triangulation with reprojection filtering.

Paths: All outputs below are read from the run directory: `outputs/img1_img2_pose_estimation_SIFT_maxfeat=4000_0.75_NNDR_thresh_4_2026-03-03_16-22-01/`
If you re-ran the pipeline and your folder name differs, update the **RUN_DIR** below in the HTML paths.

Reported Values (Staff-Provided Images)

The assignment requires reporting the random seed, intrinsics matrix *K*, SIFT feature counts, NNDR threshold, RANSAC parameters, and any staff-provided parameters modified.

Random seed	`101`
SIFT max features	`4000`

SIFT features detected	`Image 1: 3969, Image 2: 3962`
NNDR threshold	`0.75`
RANSAC iterations	`2000`
RANSAC ε	`5e-4 (squared Sampson distance, normalized coords)`
Other modified parameters	`None (used staff defaults except the reported values above)`

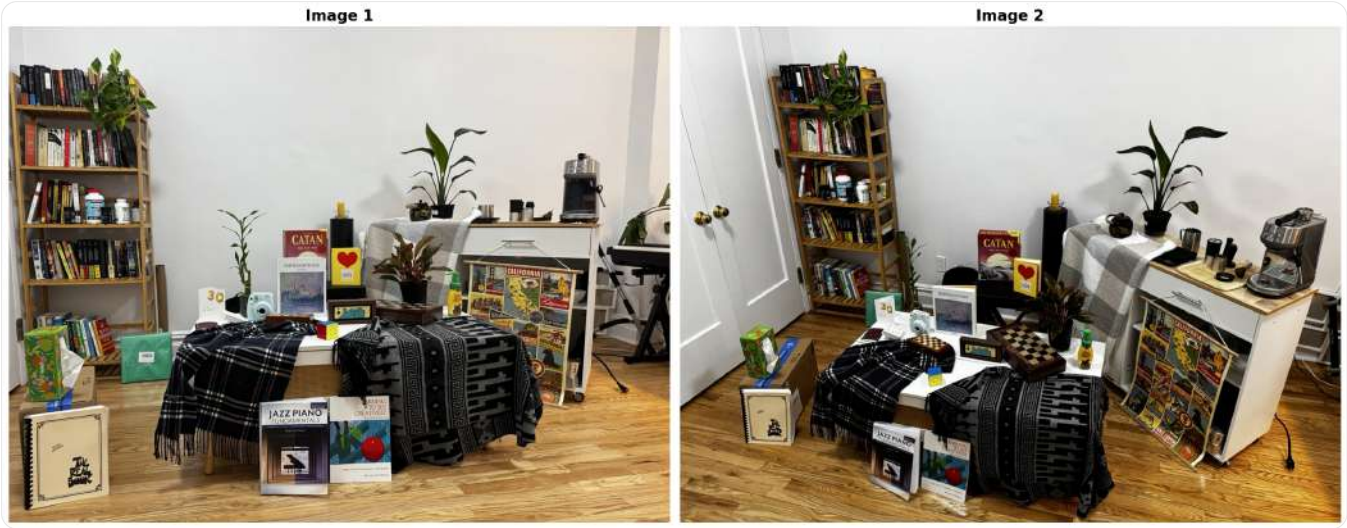
Camera Intrinsics Matrix K

```
[  887.49    0.00  640.00]
[    0.00  887.49  480.00]
[    0.00    0.00    1.00]
```

Pipeline Visualizations

The figures below correspond directly to the required deliverables: SIFT features, NNDR histogram with threshold, top-5 NNDR matches, correspondences before/after RANSAC, RANSAC convergence, epipolar lines, and final sparse reconstruction.

Step 0 — Original Images (Side-by-Side)



Original staff-provided image pair.

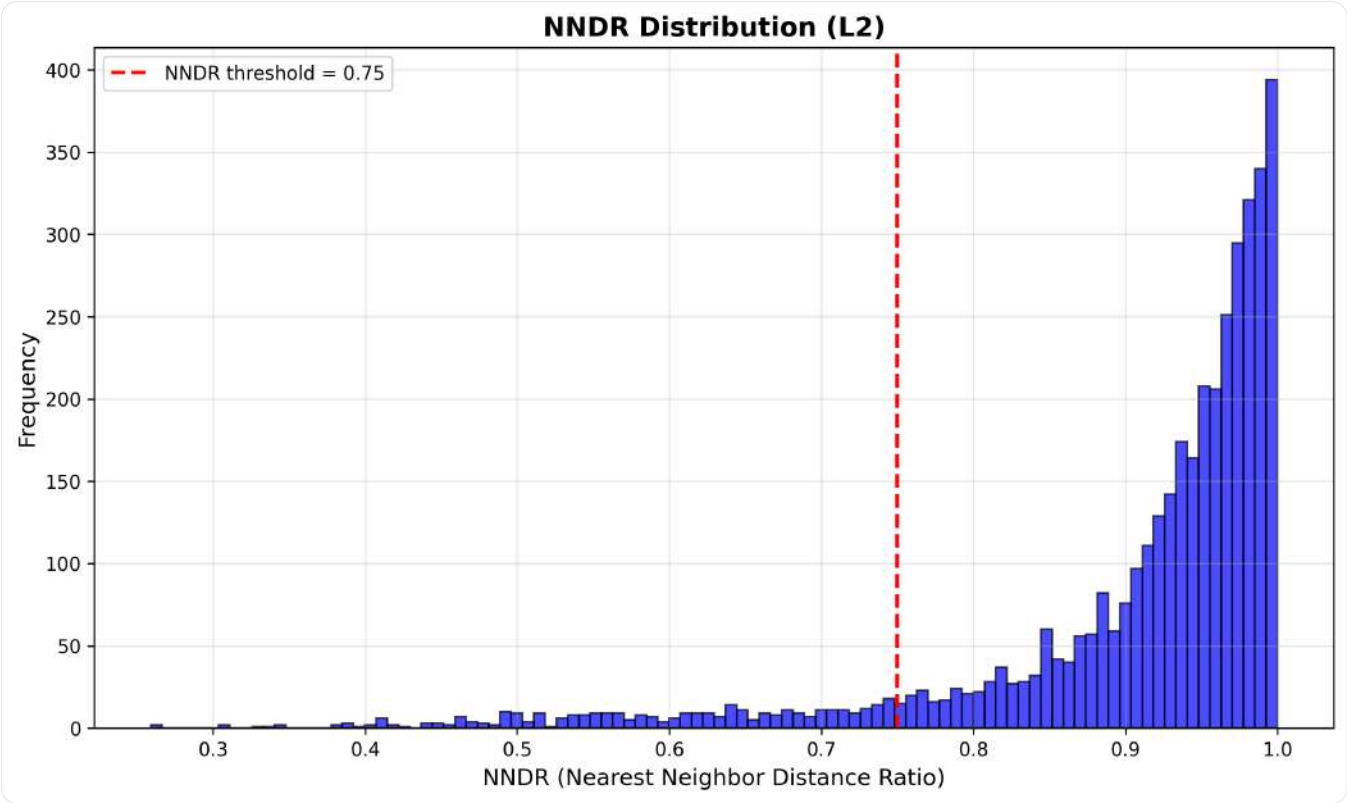
```
.../step0_original_images.png
```

Step 2 — SIFT Features (Side-by-Side)

SIFT keypoints detected in both images (max_features=4000).

```
.../step2_sift_keypoints.png
```

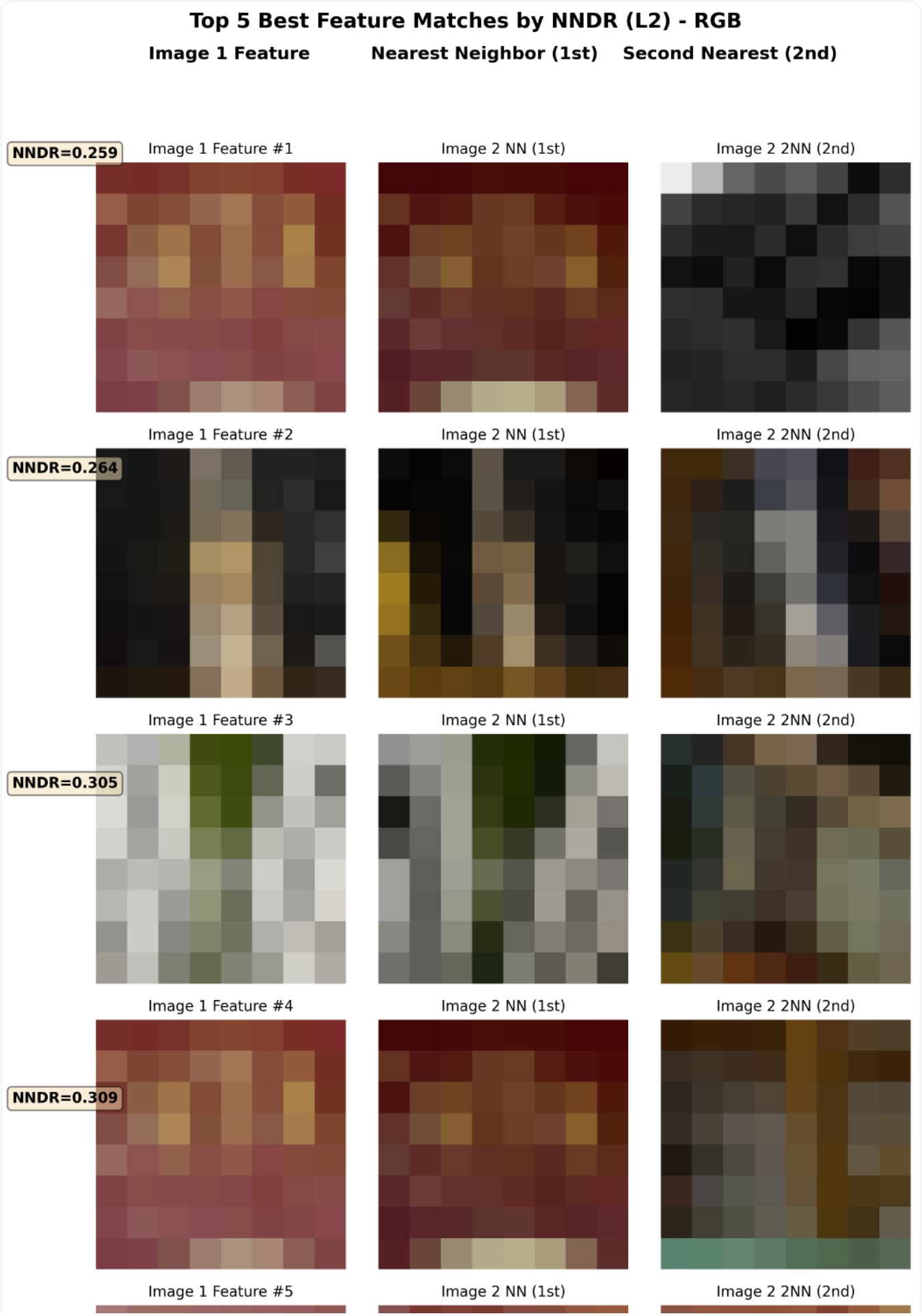
Step 3 — NNDR Histogram (Threshold Overlaid)

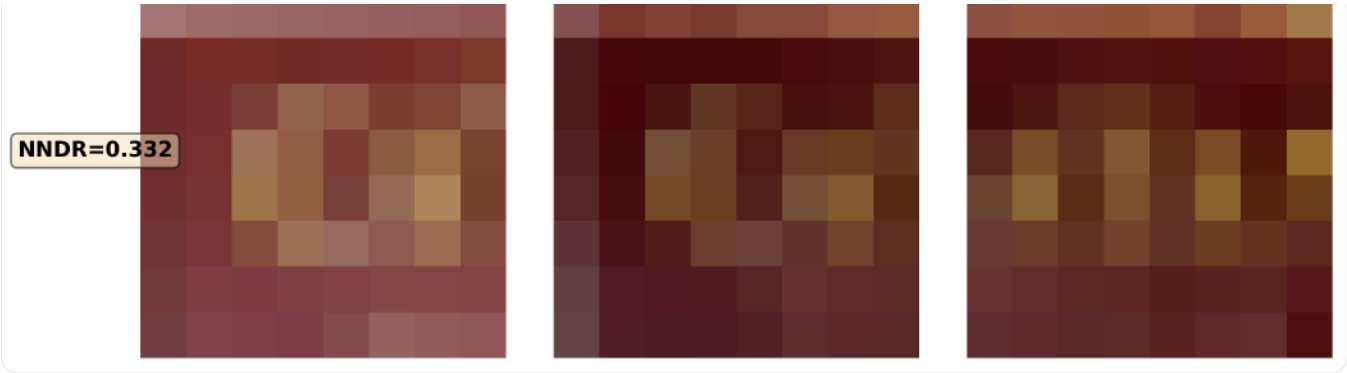


NNDR distribution (L2), threshold = 0.75.

```
.../step3_feature_matching_nndr_histogram.png
```

Step 3 — Top 5 Matches Ranked by NNDR


Top 5 Best Feature Matches by NNDR (L2) - RGB
Image 1 Feature
Nearest Neighbor (1st)
Second Nearest (2nd)
NNDR=0.259
Image 1 Feature #1
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.264
Image 1 Feature #2
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.305
Image 1 Feature #3
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.309
Image 1 Feature #4
Image 2 NN (1st)
Image 2 2NN (2nd)
Image 1 Feature #5
Image 2 NN (1st)
Image 2 2NN (2nd)



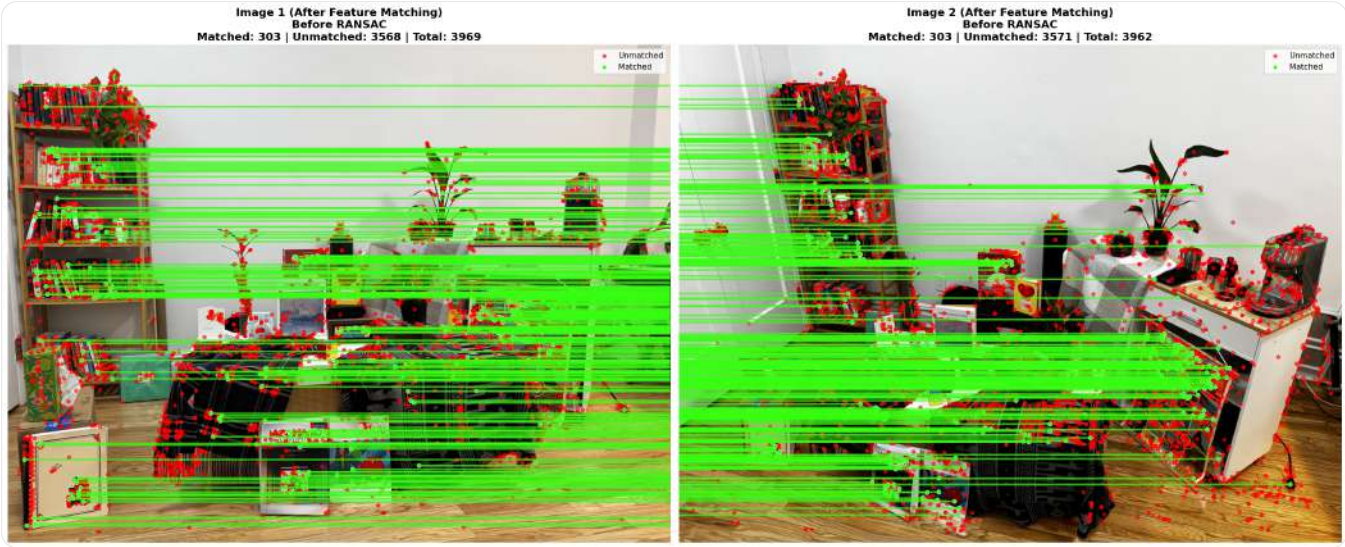
Top 5 matches ranked by NNDR.

```
.../step3_feature_matching_top5_nndr.png
```

Step 3 — Correspondences Before RANSAC

Matches before RANSAC (tentative correspondences).

```
.../step3_correspondences_before_ransac.png
```



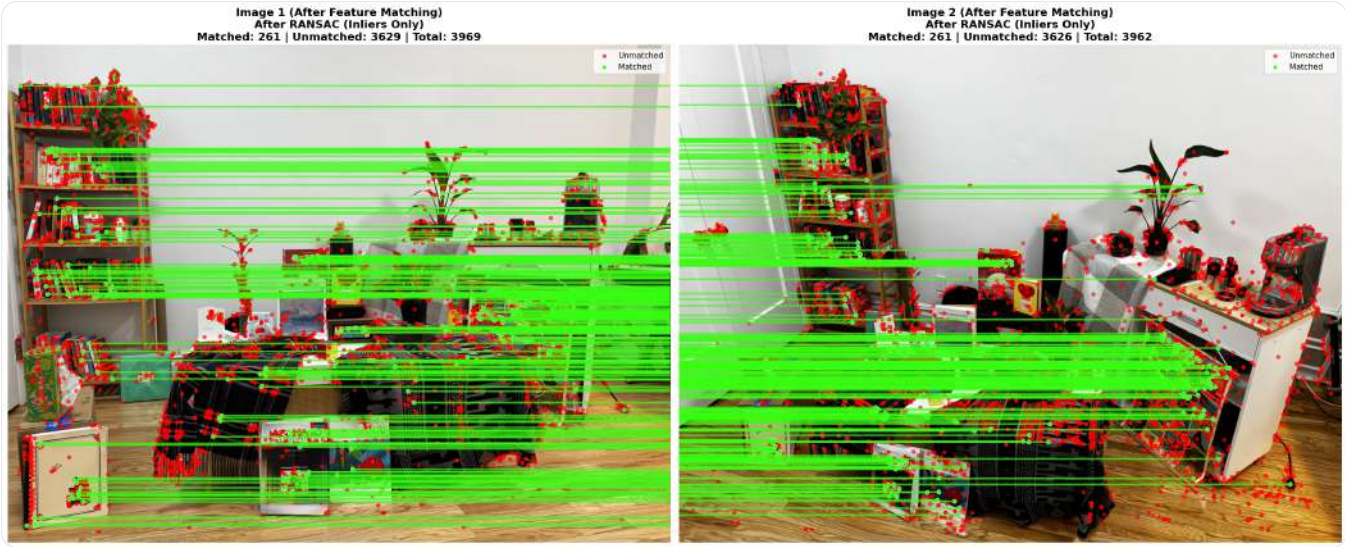
Feature matching visualization before RANSAC.

```
.../step3_feature_matching.png
```

Step 4 — Correspondences After RANSAC (Inliers)

Correspondences after RANSAC (inliers only).

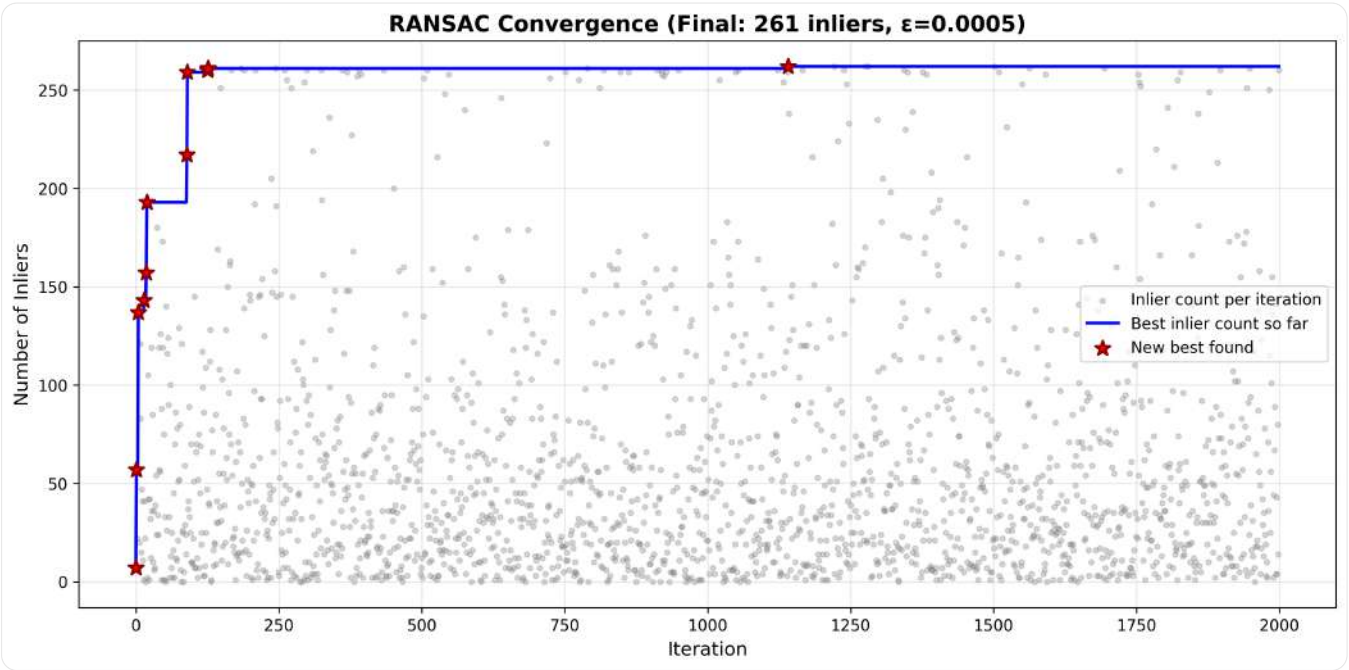
```
.../step4_correspondences_after_ransac.png
```



Feature matching visualization after RANSAC (inliers).

```
.../step4_feature_matching_after_ransac.png
```

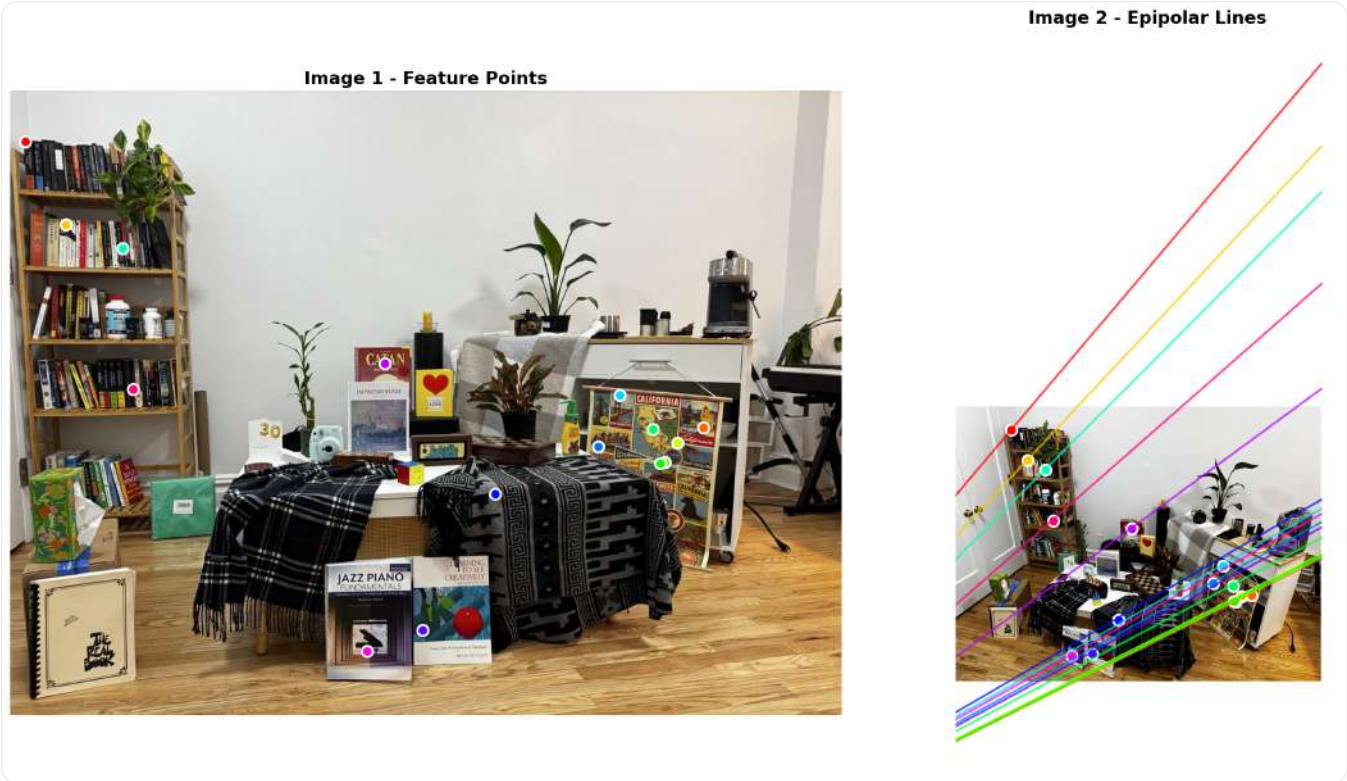
Step 4 — RANSAC Convergence



RANSAC inlier count over iterations and best-so-far curve.

```
.../step4_ransac_convergence.png
```

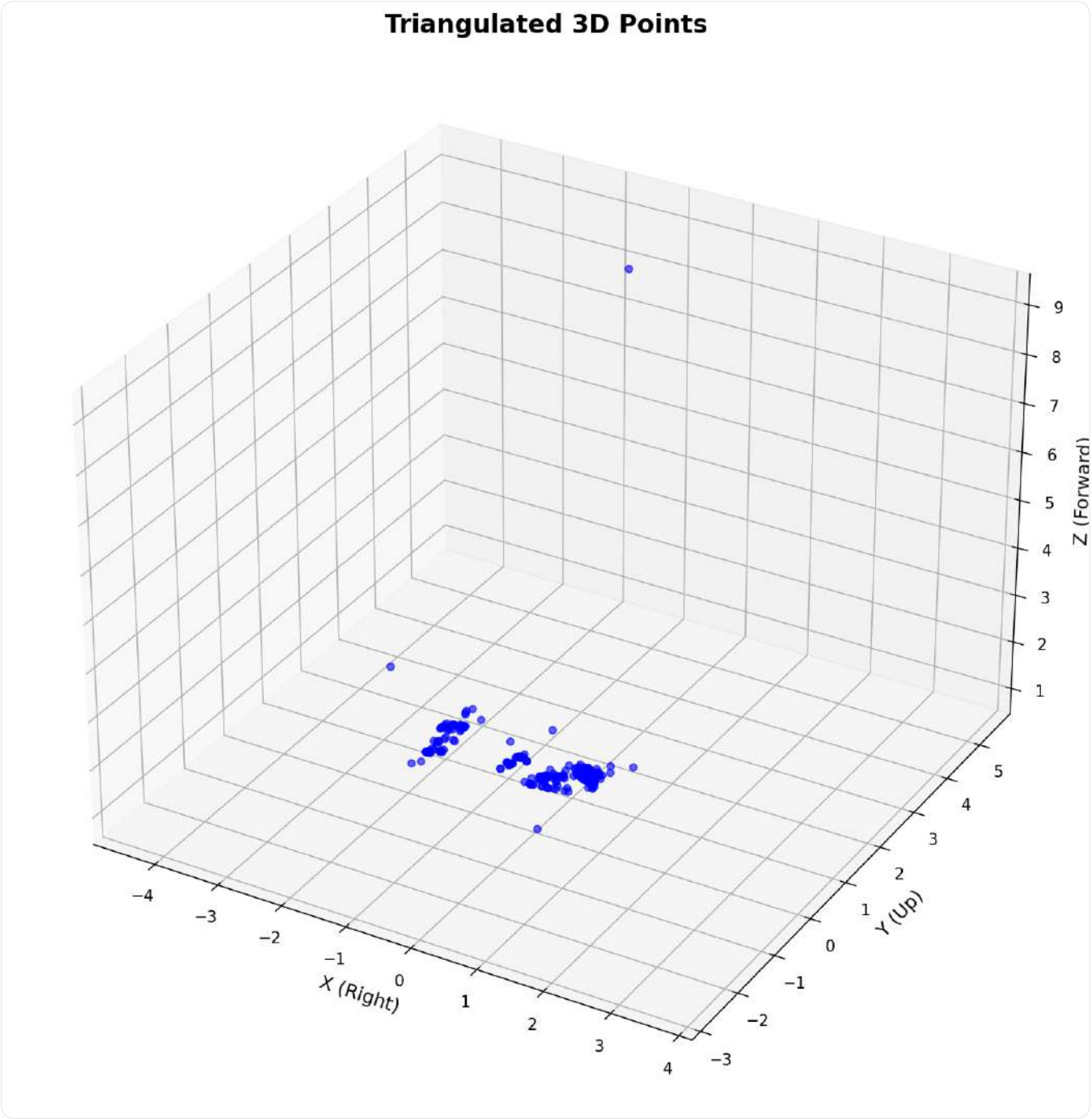
Step 4 — Epipolar Lines (Using F = K^{-T} E K^{-1})



Epipolar geometry visualization on inlier correspondences.

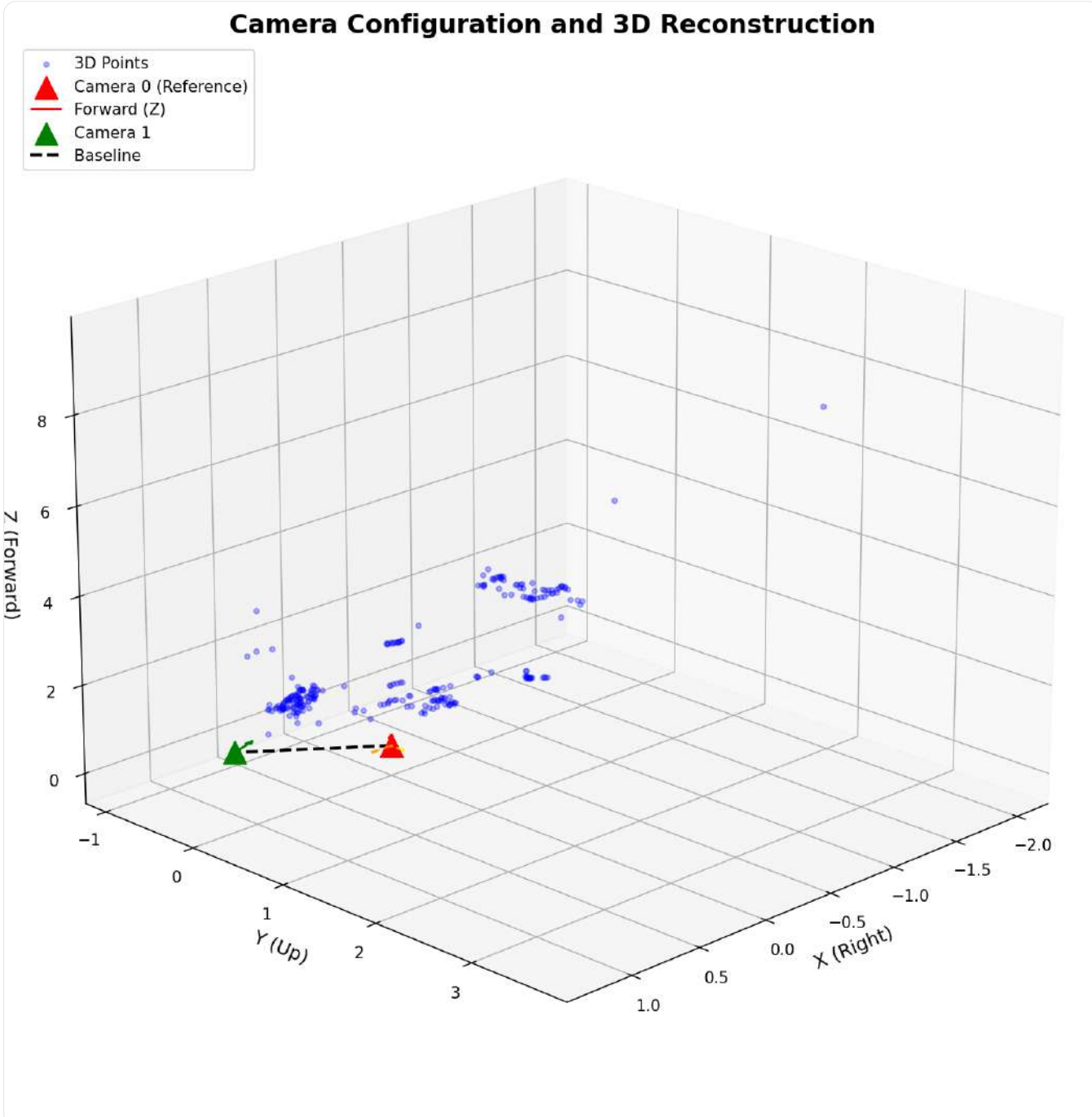
```
.../step4_epipolar_lines.png
```

Step 5 — Sparse Reconstruction



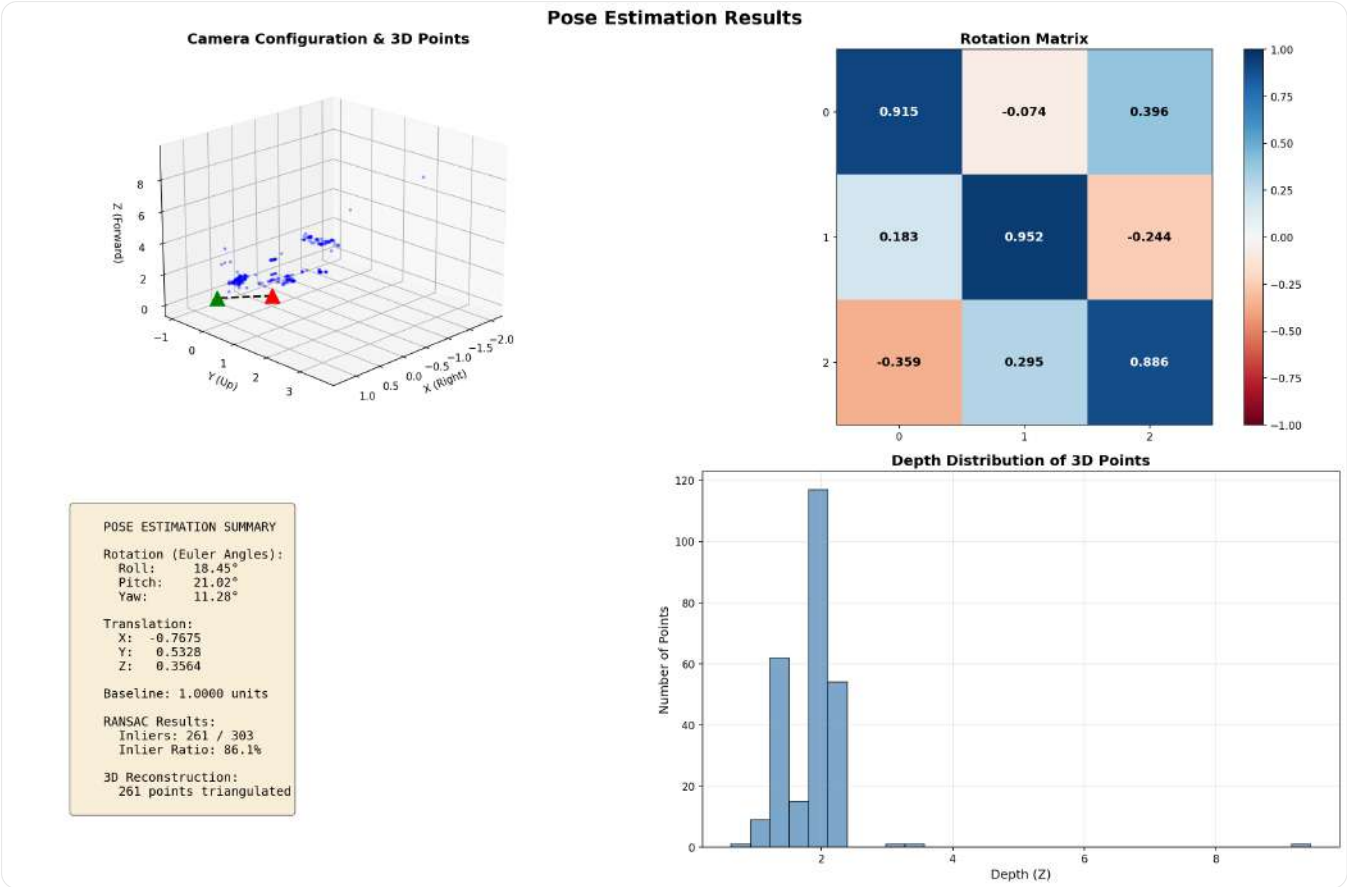
3D point cloud (sparse triangulation).

```
.../step5_3d_reconstruction.png
```



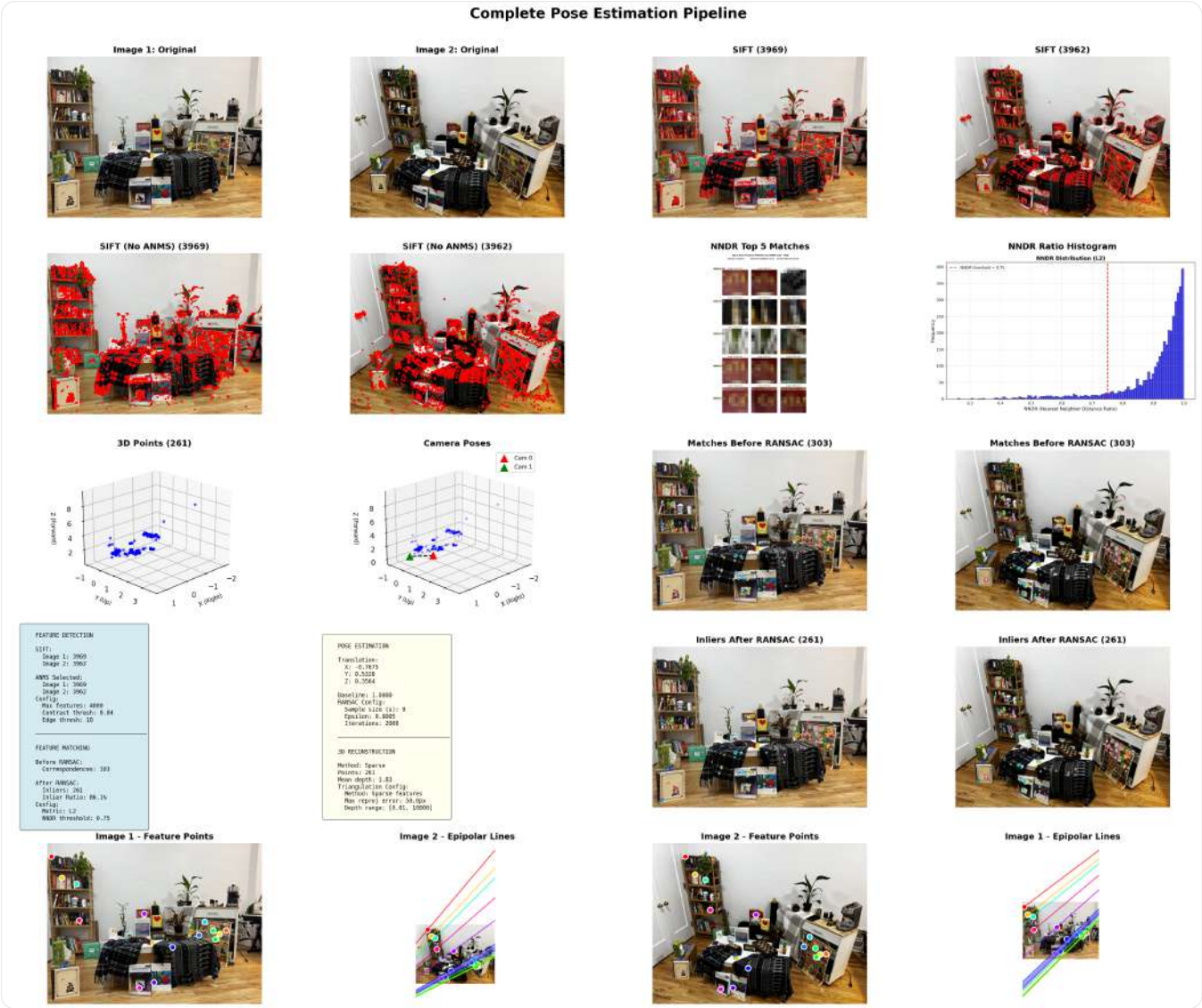
Camera configuration and 3D points.

```
.../step5_camera_poses.png
```



Pose summary (rotation/translation + depth distribution).

```
.../step5_pose_summary.png
```



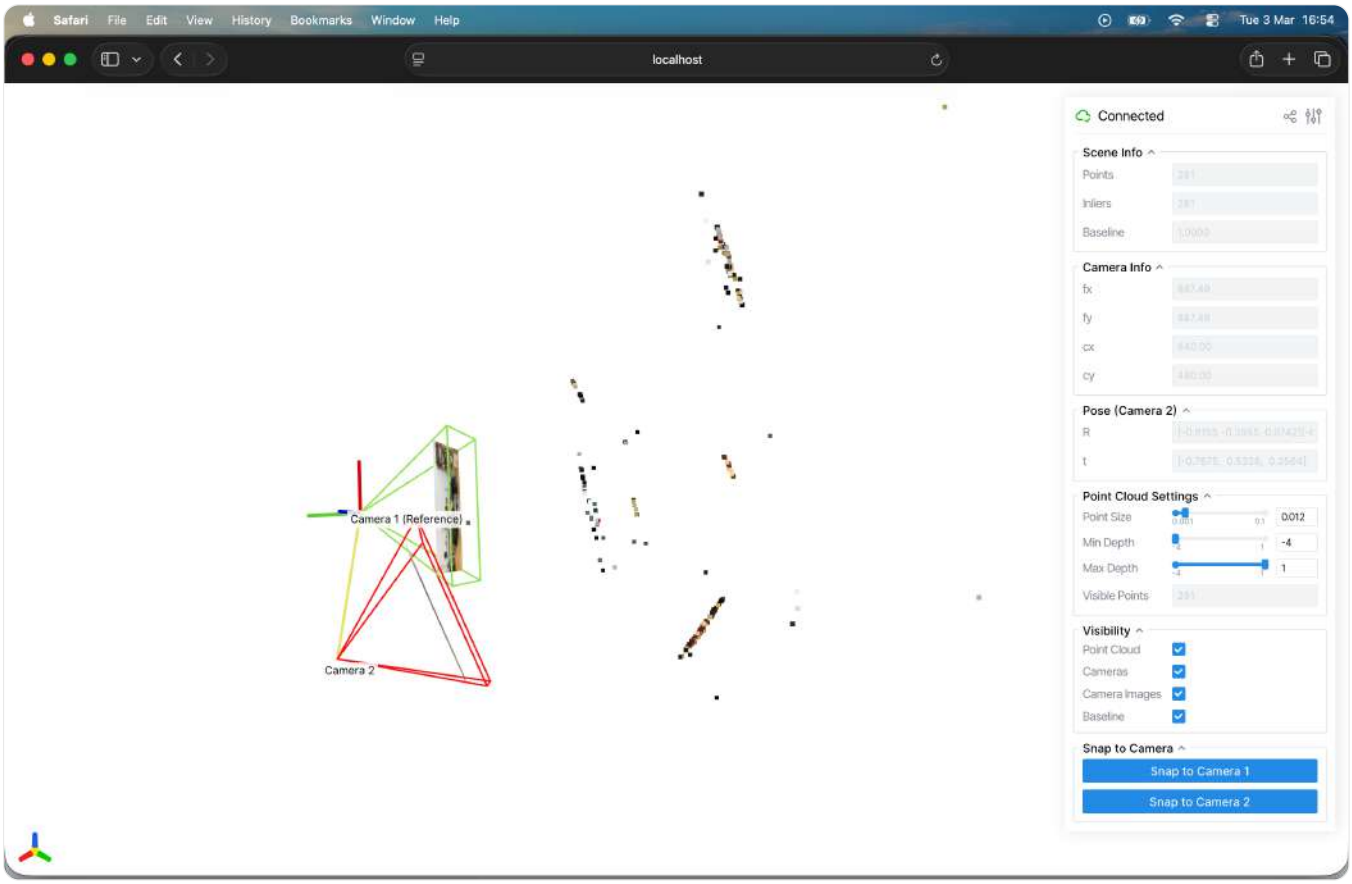
Complete pipeline grid summary.

```
.../step5_pipeline_grid.png
```

Sparse Point Cloud (viser) — Screenshots + Landmarks

Below are 3 screenshots from the viser visualization. Each screenshot includes brief landmark descriptions based on visible point clusters and the underlying scene structure.

viser Screenshot 1

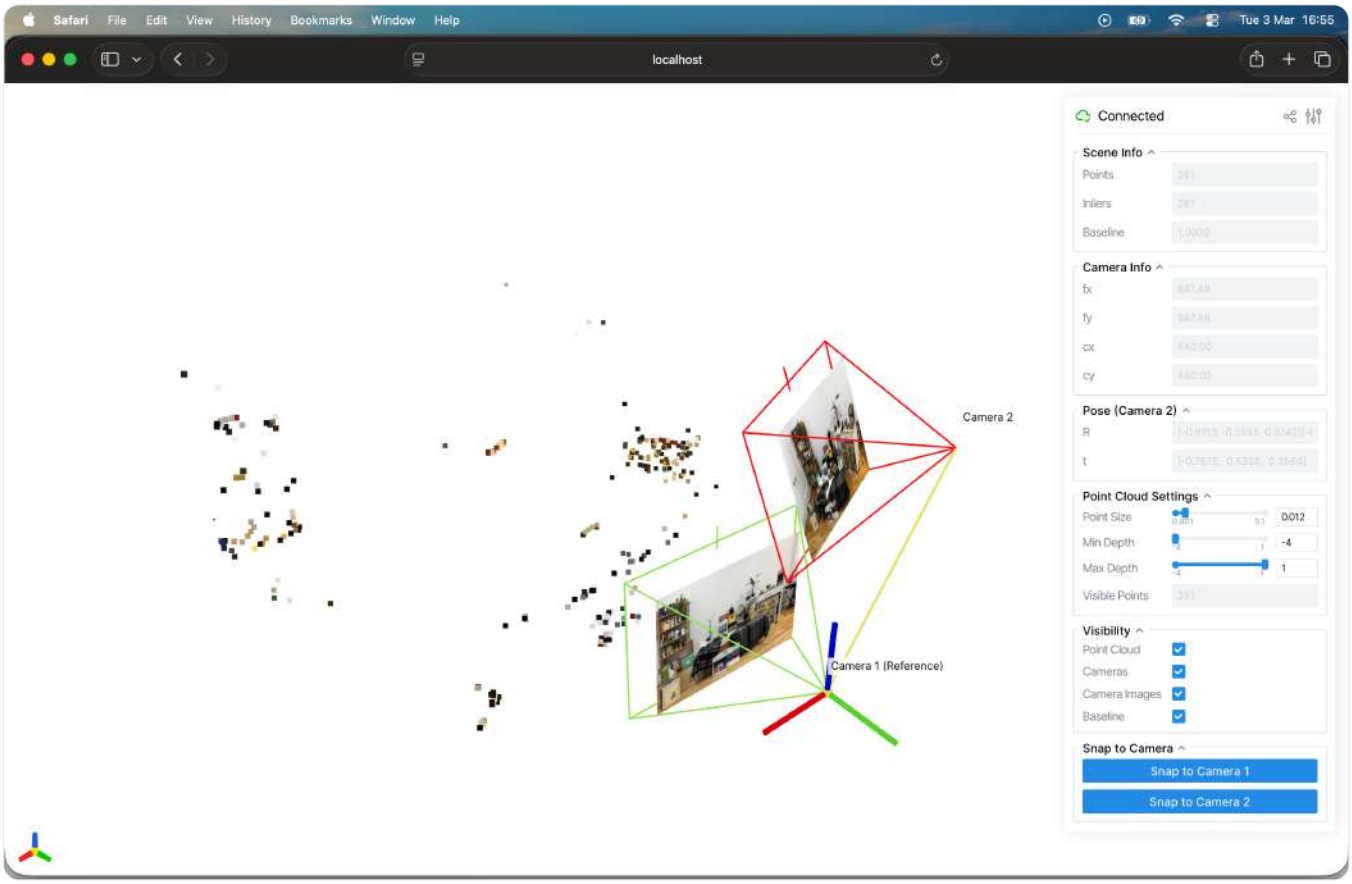


Landmark notes:

- **The top-most cluster of points** corresponds to the the bookshelf.
- **The middle cluster** corresponds to the high-texture regions on the table + objects.
- **The bottom most scattered points** correspond to poster hung on the drawer which is located toward the right side of the scene.

```
.../viser_1.png
```

viser Screenshot 2

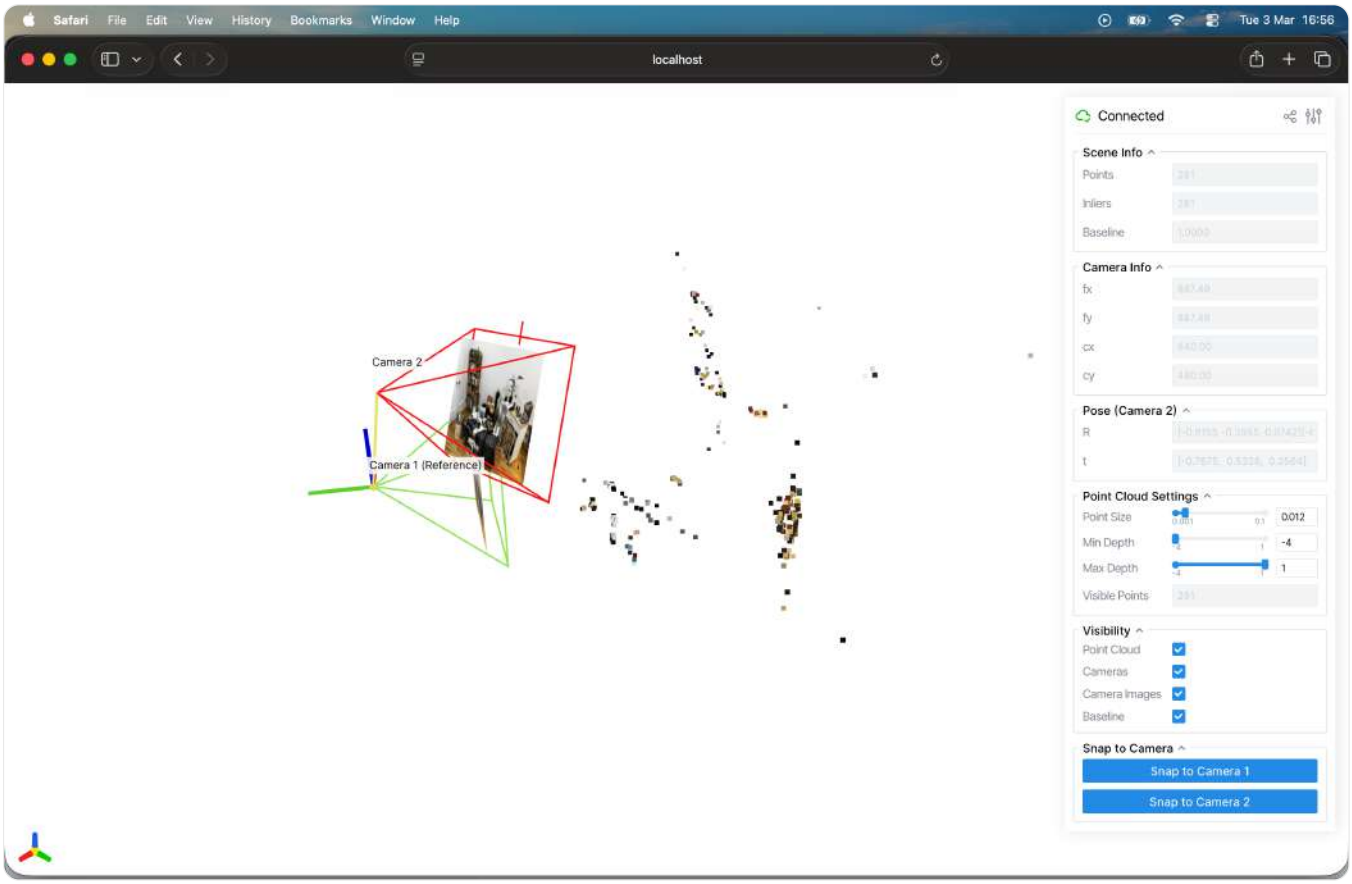


Landmark notes:

- **Left cluster** aligns with the bookshelf points and edges.
- **The middle cluster** corresponds to the objects in the very middle of the scene.
- **The points towards the upper right side** correspond to the drawer chest in the image.

```
.../viser_2.png
```

viser Screenshot 3



Landmark notes:

- **Top most cluster** represents the bookshelf / tall vertical edges.
- **Centermost points** towards the left side indicate the objects in the foreground.
- **The right most cluster** is from the stack of drawers with the poster hung from it.

`.../viser_3.png`

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HW3 — Simple Structure from Motion